QR-002 Infusion setup checklist

Doc ID	QR-002
Title	Infusion setup checklist
Parent	CS-006 §7.1; CS-006 §7.2; CS-006 §7.3; CS-006 §7.4; CS-006 §7.5; CS-006 §7.6; CS-006 §7.8; CS-006 §4; CS-008 §7
Checked against	CS-006 Rev D; CS-006 Rev D; CS-006 Rev D; CS-006 Rev D; CS-006 Rev D; CS-006 Rev D; CS-006 Rev D; CS-008 Rev E

Derived from its parent standard. If this and the parent disagree, the parent is right.

1. Infusion setup checklist, start to finish

Peel ply, then flow mesh stopping 30–50 mm short of the outlet; resin trap between bag and pump; at least 27 inHg; drop test 1 inHg or less over 10 min; only then mix IN2:AT30 at 100:30 by weight (CS-006 §7.2 to §7.5).

Also called: infusion, infuse, VARTM, resin infusion, infusion checklist, infusion setup.

1. Stack frozen on the work order; mold sealed and released with its checks passed (CS-006 §7.1).
2. Resin on hand of at least (dry fiber mass × 0.65) × 1.3, and every consumable cut before any resin is mixed (CS-006 §7.1).
3. Two people for anything bigger than a coupon, one experienced with infusion (CS-006 §7.1).
4. Plies down, peel ply over the laminate and flange edges (CS-006 §7.2).
5. Flow mesh over the whole part, stopping 30–50 mm short of the outlet side (CS-006 §7.2).
6. Feed spiral on the inlet flange, outlet spiral toward the catch pot; resin travel inlet to outlet about 0.5 m at most (CS-006 §7.2).
7. Tacky tape, bag, pleats at every corner (CS-006 §7.3).
8. Outlet line to the resin trap, then the pump; inlet line clamped shut (CS-006 §7.3).
9. Pull to 27 inHg or better (CS-006 §7.4).
10. Drop test: lose 1 inHg or less over 10 min, readings in the step note (CS-006 §7.4).
11. Mix (dry fiber mass × 0.65) × 1.2–1.3 of IN2:AT30 at 100:30 by weight (CS-006 §7.5).
12. Infuse, clamp the inlet, sign “Infuse” (CS-006 §7.5).
13. Wait the FEB hold: 48 h SLOW, 24 h FAST, from inlet clamp (CS-006 §7.6).

Each line has its own section below.

2. Before you start an infusion: what has to be ready

Before any resin is mixed: the stack is frozen on the work order, the mold is sealed and released per CS-004 with its checks passed, resin on hand is at least (dry fiber mass × 0.65) × 1.3, and every consumable is cut (CS-006 §7.1).

Also called: infusion prerequisites, prep, staging, materials, training, who can infuse.

1. Check the stack is frozen and printed on the work order (CS-006 §7.1 step 1).
2. Check the mold is sealed and released per CS-004 and its quality checks passed.
3. Weigh the cut cloth and write the total dry fiber mass on the work order (CS-006 §7.2 step 1).
4. Check there is resin on hand of at least (dry fiber mass × 0.65) × 1.3 (CS-006 §7.1 step 2).
5. Cut all consumables (peel ply, mesh, spiral, tubing, bag, tacky tape) before mixing.
6. Get a second person for any part bigger than a coupon. One of you has infused before (CS-006 §7.1 step 3).

Who can sign: “Dry stack and bag”, “Drop test, 1 inHg or less over 10 min” and “Infuse” need Resin infusion training in the app. The app stops anyone else (“this step needs Resin infusion training”). Only a lead can sign without it, and the reason goes in the work order’s event log (CS-006 §7.1 step 3).

3. Infusion stack order: peel ply, flow mesh and the mesh break

On top of the plies: peel ply over the full laminate and the flange edges, then flow mesh over the whole part, stopping 30–50 mm short of the outlet side. That gap is the mesh break: it slows resin at the end and cuts race-tracking (CS-006 §7.2 step 3).

Also called: flow mesh, infusion mesh, flow media, green mesh, mesh break, peel ply, spiral, spiral wrap, feed line, dry stack.

1. Wipe the workspace with IPA. Dust the released mold with a clean, dry microfiber only: no IPA, water or solvent on wax or PVA (CS-006 §7.2 step 1).
2. Cut cloth per the work order’s ply table.
3. Lay the plies per the work order.
4. Tack a ply with Airtac 2 only where it needs holding: can at least 30 cm (12 in) from the surface, a light mist, then wait 10 s before placing the ply (CS-006 §7.2 step 2). Heavy spray inhibits the infusion.
5. Lay peel ply over the full laminate and past the flange edges.
6. Lay flow mesh over the entire part, stopping 30–50 mm short of the outlet side.
7. Place the feed spiral along the inlet side on the flange, not on the part surface unless the work order says so (it prints through) (CS-006 §7.2 step 4).
8. Place the outlet spiral toward the catch pot.

Resin travel from inlet to outlet is about 0.5 m at most, a rule of thumb from the SN5 Composites Info & Safety primer; it varies with resin and part shape (CS-006 §7.2 step 4). A bigger part needs more feeds, planned on the work order.

4. Bagging an infusion: tacky tape, pleats and the resin trap

Tacky tape round the full perimeter, VB160 cut oversized with a pleat at every corner, and the outlet line going to the resin trap and then the pump, never bag straight to pump. The inlet line stays clamped shut (CS-006 §7.3).

Also called: bag, vacuum bag, bagging, VB160, tacky tape, sealant tape, AT-200Y, pleats, resin trap, catch pot, through-bag connector.

1. Run tacky tape around the full perimeter on the outer edge of the flange, outboard of the spirals, peel ply and mesh, with at least ½ inch between them and the tape. Keep the backing on until final placement (CS-006 §7.3 step 1).
2. Cut VB160 oversized (CS-006 §7.3 step 2).
3. Add a pleat at every corner, and every 300–400 mm of perimeter on a curved mold (CS-006 §7.3 step 2).
4. Fit the through-bag connectors (CS-006 §7.3 step 3).
5. Connect outlet line to the resin trap, and the trap to the pump.
6. Clamp the inlet line shut.

The trap is mandatory. Without it, the first overlong pull fills the vacuum pump with resin and kills it (CS-006 §4). Tape over cut spiral-wrap ends: they pierce gloves and bags alike (CS-006 §6).

5. How much vacuum (inHg) before resin, and the infusion drop test

Pull to at least 27 inHg on the team gauge. Then close the valve to isolate the pump: the bag must lose 1 inHg or less over 10 minutes. No pass, no resin. After a pass, open the valve again (CS-006 §7.4).

Also called: drop test, leak test, leak-down, vacuum level, inHg, how much vacuum, 27 inHg.

1. Pull vacuum to 27 inHg or better. Full vacuum is about 29.9 inHg at sea level (CS-006 §7.4 step 1).
2. If the gauge will not pass 25 inHg, hunt leaks now: press tacky tape seams, listen, re-seat pleats and connector bases (CS-006 §7.4 step 1).
3. Close the valve to isolate the pump. The gauge must be on the bag side of the closed valve, never on the pump (CS-006 §7.4 step 2; CS-006 §5).
4. Read the gauge, wait 10 minutes, read it again.
5. Write the start and end readings in the note on the "Drop test, 1 inHg or less over 10 min" step. It will not take a member's signature without the note, and "Infuse" cannot be signed until it is. A lead can sign without it (Sign without it), and the reason goes in the work order's event log.
6. Failed? Keep hunting (seams, pleat folds, connector bases, spiral ends piercing the bag) and test again (CS-006 §7.4 step 3).
7. Passed? Open the valve again and check the gauge climbs back to 27 inHg before you open the inlet (CS-006 §7.4 step 4).

Do not mix resin until the drop test passes, even at 11 pm with people waiting (CS-006 §7.4). A gauge that reads "psi of vacuum" is reading something else: the team's gauges read inHg (CS-006 §4). Leak hunting in detail: QR-012.

6. How much IN2 resin to mix for an infusion

Mix about (dry fiber mass × 0.65) × 1.2–1.3, at 100:30 IN2:AT30 by weight. Round up: starving an infusion mid-pull cannot be recovered (CS-006 §7.5 step 1).

Also called: resin math, how much resin, resin quantity, mix amount, batch size, IN2, AT30.

The 0.65 covers the laminate, aiming at about 60:40 fiber to resin by weight. The 1.2–1.3 covers what the mesh and the lines soak up (CS-006 §7.5 step 1). Have at least (dry fiber mass × 0.65) × 1.3 on hand before you start (CS-006 §7.1 step 2).

1. Take the dry fiber mass you wrote on the work order.
2. Multiply by 0.65, then by 1.2 to 1.3. Round up.
3. Weigh IN2 and AT30 at 100:30 by weight (CS-006 §7.5 step 2).
4. Mix and double-pot per CS-008: at least 2 min, then a second cup and a fresh stick.
5. Record resin mixed against resin used on the work order, so the factor can be tuned.

Pot life: with SLOW, plan on 80–100 min. That is the TDS text, which gives no temperature; its table says 120–150 min at 25 °C (CS-006 §7.5 step 2). On a long infusion you may mix the total as two or more smaller batches and top up the pot as it draws down, keeping the inlet submerged (CS-006 §7.5 step 1).

7. Running the infusion: opening the inlet to clamping off

With 27 inHg or better confirmed and the drop test passed, open the inlet clamp slowly and watch the flow front. When resin reaches the outlet spiral and the laminate is fully dark, clamp the inlet and leave the pump running through gel (CS-006 §7.5).

Also called: infuse, infusing, flow front, open the inlet, clamp off, wet out.

1. Brace the resin cup (CS-006 §7.5 step 3).
2. Submerge the inlet line end fully. A slurping inlet injects bubbles.
3. Confirm the valve is open again after the drop test and the gauge reads 27 inHg or better (CS-006 §7.4 step 4; §7.5 step 4).
4. Open the inlet clamp slowly.
5. Watch the flow front for even progress and no race-tracking. A larger part takes about 20–30 min, sometimes longer (CS-006 §7.5 step 4).
6. When resin reaches the outlet spiral and the laminate is fully dark and wet, clamp the inlet (CS-006 §7.5 step 5).
7. Leave the pump running through gel.

8. Write the times (mix, open, full wet, clamp), resin mixed against used, gauge readings and anything odd, with photos, on the work order (CS-006 §7.5 step 6).
9. Sign "Infuse". Set "Resin mixed" to IN2 + AT30 SLOW or IN2 + AT30 FAST, set "Finished on" and "at" to the time you clamped the inlet, and fill in "Shop temperature, °C (optional)".

That clamp time starts the hold on "Cure and demould". Problems mid-infusion (race-tracking, a dry patch, bubbles): QR-012.

8. Do we degas the resin? (degassing IN2)

Sometimes. Degassing is optional: the team degasses IN2 sometimes, and the IN2 datasheet does not call for it (CS-006 §7.8). Degas when the part shows a visible cosmetic face, after pinholes or bubbles from the same mold, or when the mix looks cloudy.

Also called: degas, degassing, de-gas, vacuum chamber, bell jar, pinholes, bubbles in resin.

How (CS-006 §7.8):

1. Mix and double-pot first (CS-008 §7 step 3).
2. Put the mixed cup in the degassing chamber.
3. Pull vacuum until the foam rises and collapses.
4. Infuse straight away.

Which chamber, what vacuum level and how long under the bell are not written down yet: CS-006 §7.8 marks them for Nick to confirm. Ask a lead.

What to watch (CS-006 §7.8):

- The foam rises several times the resin's height before it collapses. Leave headroom in the cup, or it boils over into the chamber.
- Time under the bell comes out of the pot life: plan on 80–100 min for SLOW and 9–14 min for FAST (IN2 TDS text, CS-006 §7.8). A FAST mix can spend most of its pot life under the bell, so check the clock first.
- A warm cup means the pot life is nearly spent. Do not infuse it; treat it as a hot cup per CS-008 §6.

Degassed or not, keep air out: mix without whipping air in, keep the inlet submerged, hold 27 inHg or better, pass the drop test, and note on the work order where any bubbles showed and whether the resin was degassed (CS-006 §7.8).

9. When can I demould (demold) an infusion?

48 h for IN2 + AT30 SLOW and 24 h for IN2 + AT30 FAST, counted from the time you clamped the inlet. Those are the team's FEB holds (CS-006 §7.6 step 1; CS-008 §5.1). The datasheet demould figure is shorter: 18–24 h SLOW and 6–8 h FAST at 25 °C.

Also called: demould, demold, de-mould, cure hold, FEB hold, cure time, pull the part.

"Cure and demould" on the work order stays locked until the hold passes. Only a lead can override it, with a written reason (CS-006 §7.6 step 1). A colder shop runs longer than either figure, and gel is not cure.

1. Wait out the hold.
2. Demould (CS-006 §7.6 step 3).
3. Label the part per CS-001.
4. Photograph it, weigh it, and record both in the work order's quality checks.
5. Hand it on to trimming (CS-009).
6. On the Molds tab, open the mold, add 1 to Parts pulled and set Stage back to Sealed: the PVA came off with the part. It goes back to Ready for layup only after the next release (CS-006 §7.6 step 4).

Full mechanical properties take at least 14 days (CS-006 §7.6 step 2). Trimming can be scheduled freely, but avoid structural or fit loading before then. Every resin's hold and the reasons behind it: QR-001.